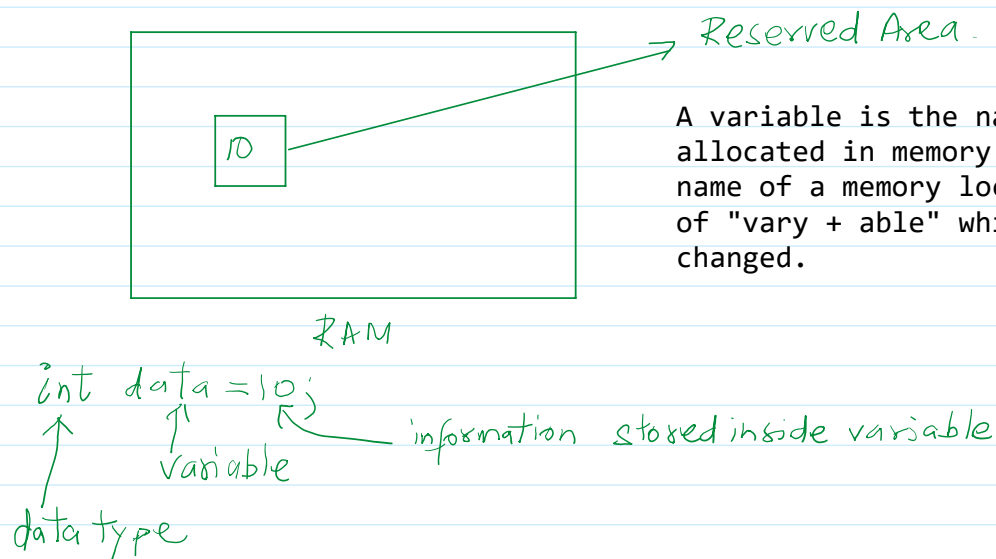


Java Variables, Data Types and Operators

14 September 2026 06:30



Types of variables:

There are 3 types of variables in Java:

- o Java local variable
- o Java Instance variable
- o Java Static variable

Local Variable: A variable declared inside the body of the method is called local variable. You can use this variable only within that method and the other method in the class and even aware of that variable exist.

A local variable cannot be defined with "static" keyword.

Example:

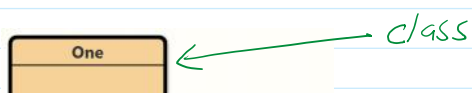
```
int x = 10;
System.out.print("\nx = " + x);
```

Output:

x = 10

Java Instance variable: A variable declared inside the class but outside the body of the method is called an instance variable. It is not declared as static.

It is called an instance variable because its value is in instance-specific and is not shared among instances.



class

keyword use to create instance of a class

constructor method
name is same as class
it won't have any return type.
it is created by JVM. (Java Virtual Machine)

Java Compiler + Interpreter

or instance

BlueJ: Create Object

One()

Name of Instance: first

OK Cancel

class bench or package bench

Object or instance bench

code pad
JShell

object / instance are used to call methods present in class.

JVM indicator.

reset JVM button.

Creating object...

Code editor

One X

Class Edit Tools Options

Compile Undo Cut Copy Paste Find... Close

Source Code

```
1 package Chapter1;
2
```

One - ProjectOne

Code editor

Class Edit Tools Options

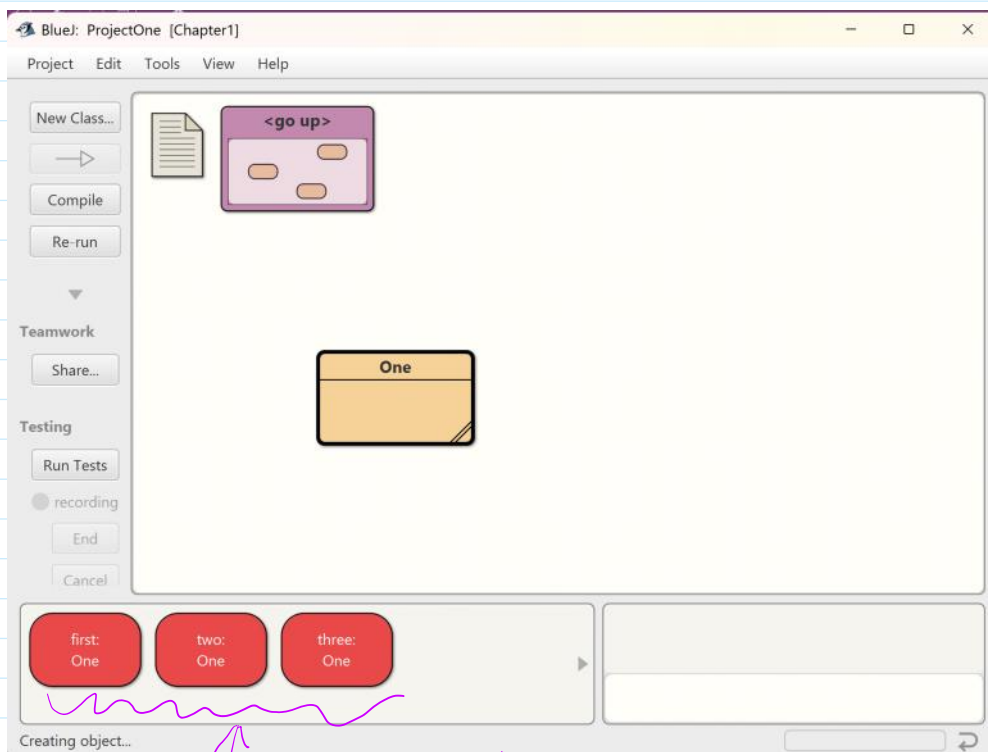
One X

Compile Undo Cut Copy Paste Find... Close Source Code

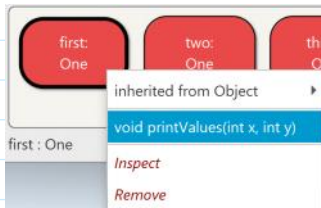
```
1 package Chapter1;
2
3
4 /**
5  * Write a description of class One here.
6  *
7  * @author (your name)
8  * @version (a version number or a date)
9  */
10 public class One
11 {
12     public static void main(String[] args){
13
14     }
15     /**
16     * Input integer value in x
17     * Input integer value in y
18     */
19     public void printValues(int x , int y){
20         System.out.print("\nx = " + x);
21         System.out.print("\ny = " + y);
22     }
23 }
```

Class compiled - no syntax errors saved

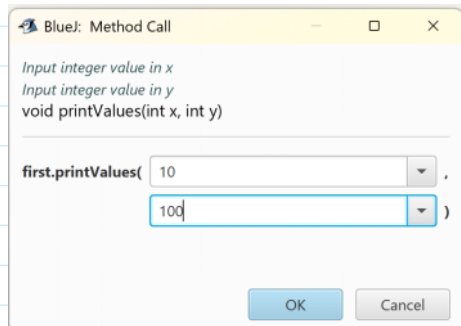
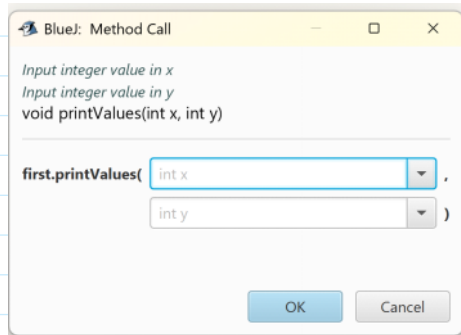
Msg box



We can create multiple objects or instances.



← This method is present in my class, we can call or invoke this method using the object or instance.



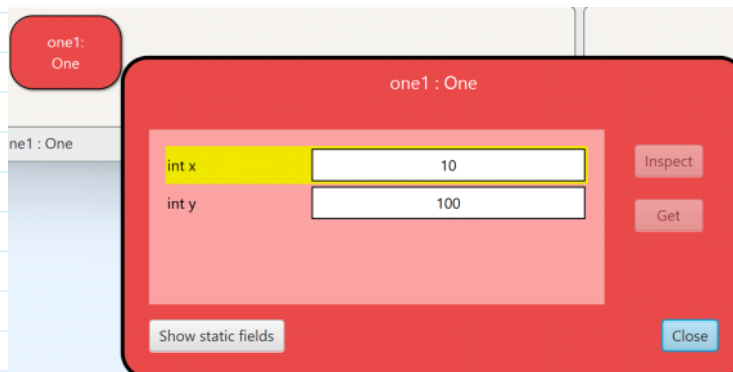
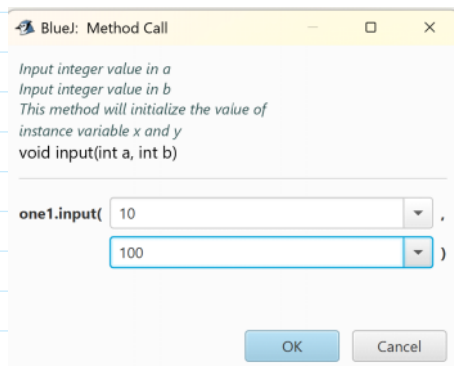
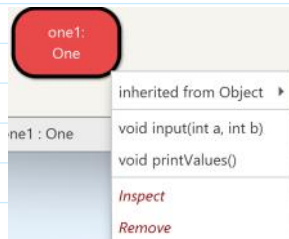
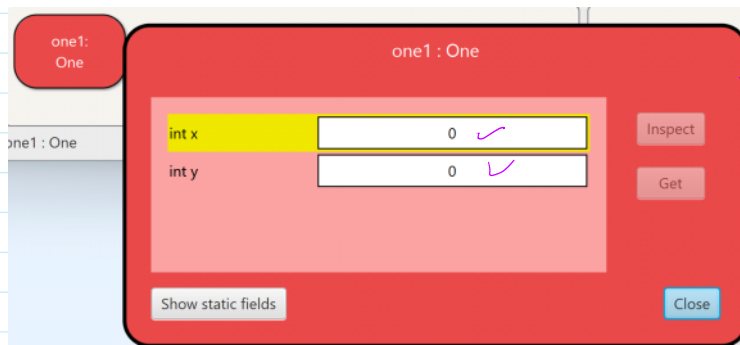
x = 10
y = 100 } ← System.out

System.out.^{printf} print(
println)

← The terminal window is divided into 3-area

Can only enter input while your program is running

← System.in



```
package Chapter1;
```

```
/**
 * Write a description of class One here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class One
{
    //instance variables
```

```

int x;
int y;
public static void main(String[] args){

}
/**
 * Input integer value in a
 * Input integer value in b
 * This method will initialize the value of
 * instance variable x and y
 */

public void input(int a, int b){
    x = a;
    y = b;
}

public void printValues(){
    System.out.print("\nx = " + x);
    System.out.print("\ny = " + y);
}
}

```

"this" keyword in Java: It is used to initialize instance variables with the value of local variables or parameter variables.

```

public class Student

```

```

{
    int rollNum;
    String name;

    Student(int rn, String nm){
        this.rollNum = rn;
        this.name = nm;
    }

```

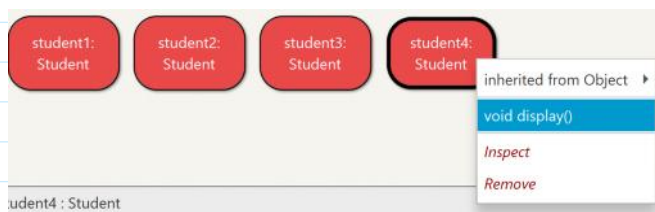
parameters, they take data input while creating an object.

```

    void display(){
        System.out.print("\nRoll number: " + rollNum);
        System.out.print("\nName: " + name);
    }
}

```

We have written this method only once, but called it multiple time with multiple objects.



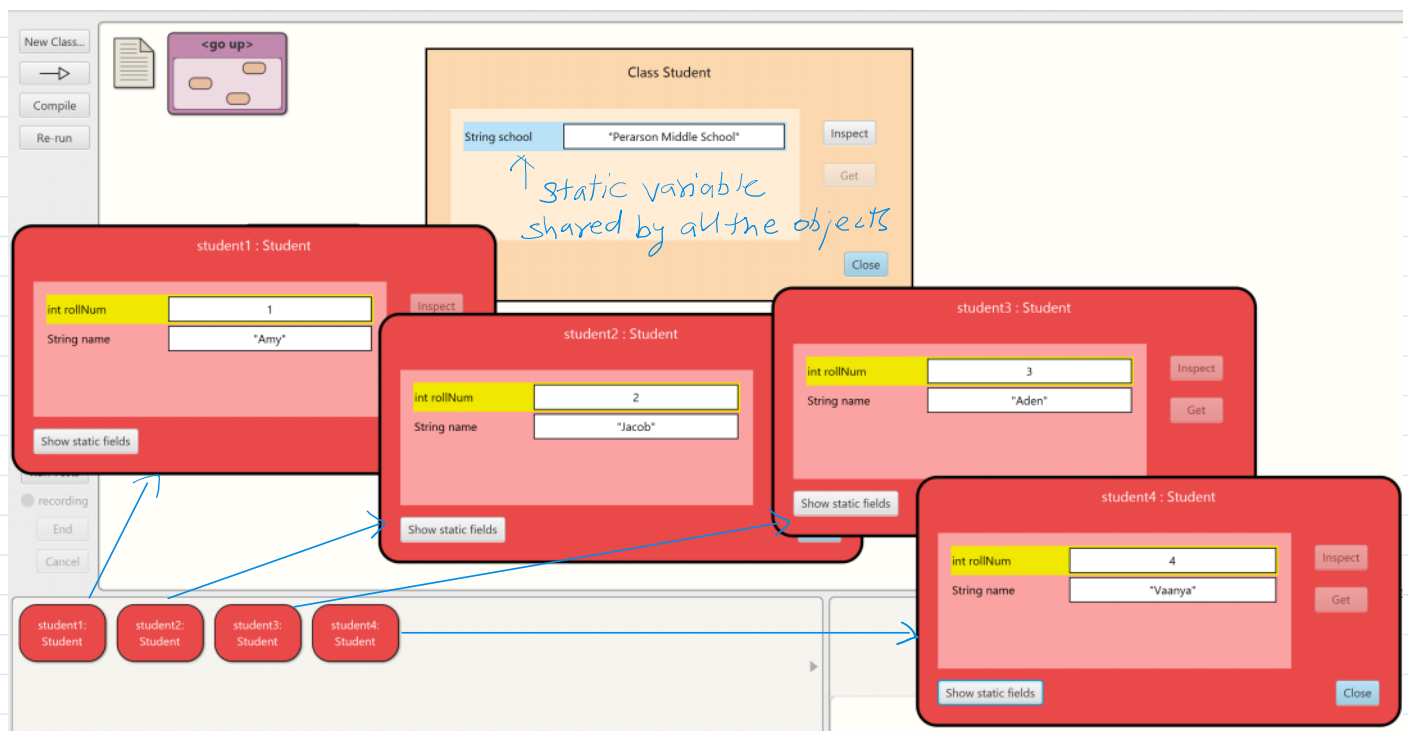
This method gets associated with all the objects that we create and works by using the instance variables of that object.

```
BlueJ: Terminal Window - ProjectOne
Options

Roll number: 4
Name: Anchit
Roll number: 3
Name: Keshav
Roll number: 2
Name: Vaanya
Roll number: 1
Name: Jiya
```

Static Variable

A variable that is declared as static is called a static variable. It cannot be local. You can create a single copy of a static variable and share it among all the objects of the class. Memory allocation of static variable happens only once when the class is loaded in the memory.



```
package Chapter1;
```

```
/**
 * Write a description of class Student here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Student
{
    int rollNum;
    String name;
```

```
//static is shared by all the objects or instances of the class.
static String school = "Perarson Middle School";

Student(int rn, String nm){
    this.rollNum = rn;
    this.name = nm;
}

void display(){
    System.out.print("\nRoll number: " + rollNum);
    System.out.print("\nName: " + name);
    System.out.print("\nSchool: " + school);
}
}
```

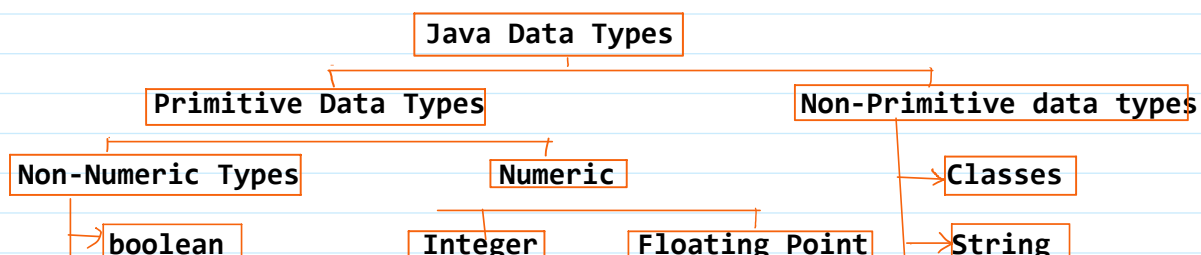
Output:

```
Blue: Terminal Window - ProjectOne
Options
Roll number: 1
Name: Amy
School: Perarson Middle School
Roll number: 2
Name: Jacob
School: Perarson Middle School
Roll number: 3
Name: Aden
School: Perarson Middle School
Roll number: 4
Name: Vaanya
School: Perarson Middle School
```

Data types in Java

In programming languages, data types specify the different sizes(bits or bytes) and values that can be stored in the variable or constants. Each data type is pre-defined, which makes Java a statically and strongly typed language. There are following two types of data types in Java.

- 1. Primitive Data Types:** The primitive data type include boolean, byte, short, int, long float, double, char.
- 2. Non-Primitive Types:** The non primitive data type include Classes, Interfaces, String and Arrays.



Non-Numeric Types

- boolean
- char

Numeric

- Integer
 - byte
 - short
 - int
 - long
- Floating Point
 - float
 - double

Classes

- String
- Arrays
- Objects
- Interfaces
- Enums

Hierarchy of data types in Java

Primitive Data Types

Data-type	Size	Range of values
boolean	1-bit	true or false
char	2 bytes	Unicode values
byte	1 byte	-128 to 127
short	2 byte	-32768 to 32767
int	4 byte	-2147483648 to 2147483647
long	8 bytes	-9223372036854775808 to 9223372036854775807
float	4 bytes	It support 6 decimal place approximation
double	8 bytes	It supports 15 decimal place approximation